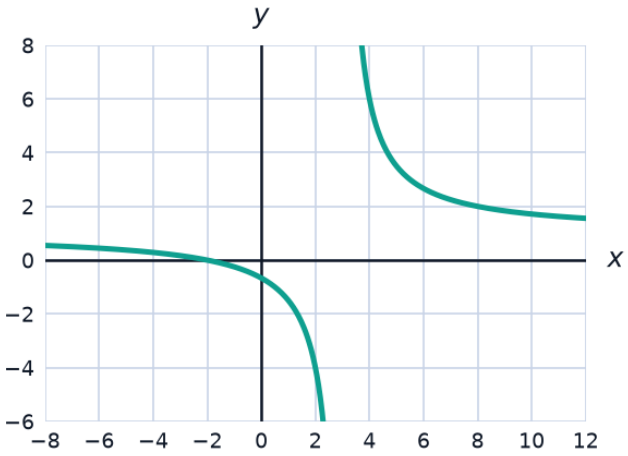
Graphing Rational Functions

ClipSAT Printable Practice Suite (clipsat.org) • Topic 8.02

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

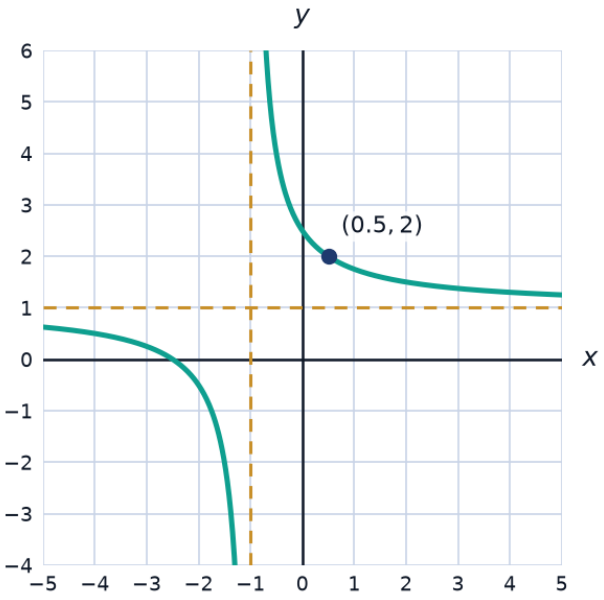
1 [Foundational] • Target: Reading graphs

The graph of $f(x) = \frac{x+2}{x-3}$ is shown. From the graph and the formula, state the asymptotes and intercepts.



2 [Foundational] • Target: Reading graphs

The function $g(x) = \frac{1}{x-2} + 3$ is a transformation of $y = \frac{1}{x}$.



a Describe the transformation.

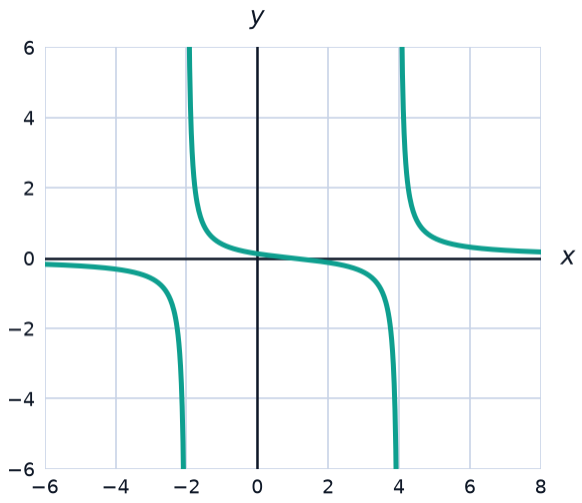
b State the asymptotes of g.

c State the domain and range.

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3 [Foundational] • Target: Sign analysis

Let $f(x) = \frac{x-1}{(x+2)(x-4)}$. Using a sign chart with critical values $x = -2, 1, 4$, determine the intervals on which $f(x) > 0$.



4 [Foundational] • Target: Sign analysis

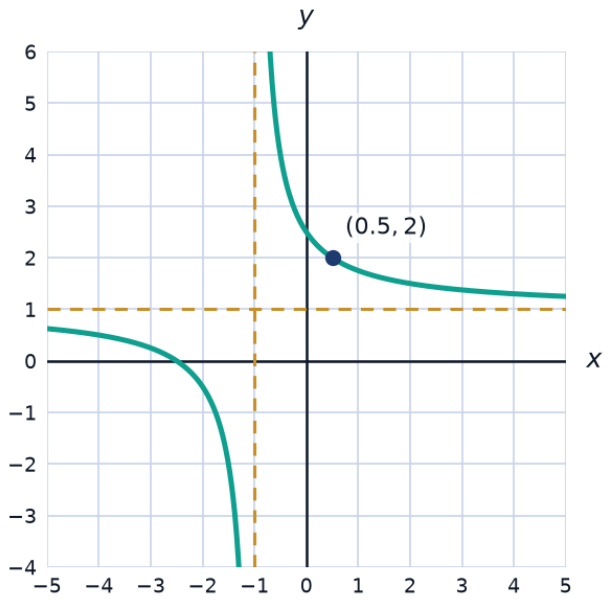
Solve the rational inequality $\frac{x+3}{x-2} \le 0$.

5 [Foundational] • Target: Sign analysis

Sketch-planning: for $h(x) = \frac{2x}{x^2-1}$, state the asymptotes and the symmetry of the graph.

6 [Foundational] • Target: Sketching a complete graph

For $f(x) = \frac{x^2-1}{x^2-4}$, find all intercepts and asymptotes, then describe the graph's behaviour near each vertical asymptote.



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7 [Foundational] • Target: Sketching a complete graph

Sketch-planning: the function $g(x) = \frac{3}{x-1} - 2$ has a vertical asymptote at $x = 1$ and horizontal asymptote $y = -2$. Find its x- and y-intercepts.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: More rational inequalities

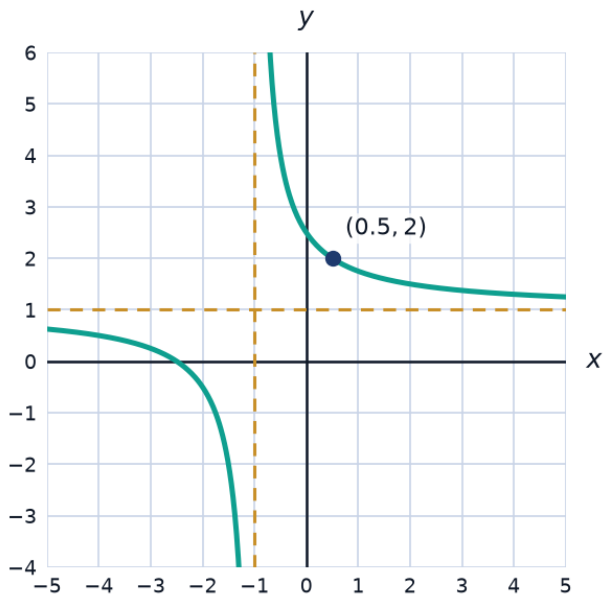
Solve $\frac{(x-2)(x+1)}{x-4} \geq 0$.

9 [Intermediate] • Target: More rational inequalities

Solve $\frac{2}{x+3} > 1$.

10 [Intermediate] • Target: Building a rational function from a description

Construct a rational function with vertical asymptote $x = 2$, horizontal asymptote $y = 1$, and a hole at $x = -3$.



11 [Intermediate] • Target: Graphing with a slant asymptote

For $f(x) = \frac{x^2-4}{x-1}$, find the slant asymptote and the vertical asymptote, and use them together with intercepts to describe the overall shape of the graph.

12 [Intermediate] • Target: Full graph analysis: putting it all together

For $f(x) = \frac{2x^2-2}{x^2-9}$, find all intercepts, all asymptotes, and describe the symmetry of the graph (even, odd, or neither).

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13 [Intermediate] • Target: A compound rational inequality

Solve $\frac{x^2-1}{x-2} \leq 0$.

14 [Intermediate] • Target: Graphing near a hole

For $f(x) = \frac{x^2-9}{x-3}$, describe how the graph looks immediately around $x = 3$, and find the coordinates of the hole.

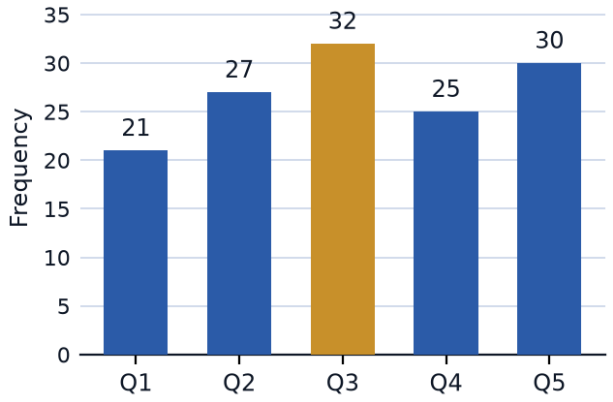
Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Graphing near a hole • Cross-Lesson: 3.04 Double-Angle Formulas and Trigonometric Equations

Solve $\frac{3}{x+1} \geq \frac{2}{x-1}$.

16 [Challenging] • Target: Graphing near a hole • Cross-Lesson: 4.01 Circles and Parabolas

A rectangular garden's cost per square meter to fence is modeled by $C(x) = \frac{200}{x} + 8x$, where x is the garden's width in meters. Find the vertical asymptote and horizontal/slant asymptote of this function, and interpret what the vertical asymptote means physically (why $x = 0$ isn't a realistic input).



17 [Challenging] • Target: Graphing near a hole • Cross-Lesson: 4.02 Ellipses

State the domain of $f(x) = \frac{\sqrt{x}}{x-4}$.

18 [Challenging] • Target: Building a rational function from a description • Cross-Lesson: 4.03 Hyperbolas

For $f(x) = \frac{(x+2)(x+1)}{(x+1)(x+3)}$, identify the hole and vertical asymptote.

19 [Challenging] • Target: Graphing with a slant asymptote • Cross-Lesson: 5.01 Vectors: Components and Operations

For $f(x) = \frac{(x+1)(x+2)}{(x+2)(x+5)}$, identify the hole and vertical asymptote.

20 [Challenging] • Target: Full graph analysis: putting it all together • Cross-Lesson: 5.02 The Dot Product

For $f(x) = \frac{(x-0)(x+1)}{(x+1)(x+3)}$, identify the hole and vertical asymptote.

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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiplicity of Zeros and Sign Analysis

If a factor $(x - c)$ in the denominator of a rational function has odd multiplicity (e.g. $(x - c)^1$), how does the graph behave near the asymptote $x = c$?

a. The graph goes in opposite directions ($+\infty$ on one side and $-\infty$ on the other)

b. The graph goes in the same direction ($+\infty$ on both sides)

c. The graph crosses the axis

d. A hole is formed

22 [Foundational] • Target: Multiplicity of Zeros and Sign Analysis

If the factor in the denominator has even multiplicity $(x - c)^2$, how does the graph behave near $x = c$?

a. The graph goes in the same direction on both sides (both $+\infty$ or both $-\infty$)

b. The graph goes in opposite directions

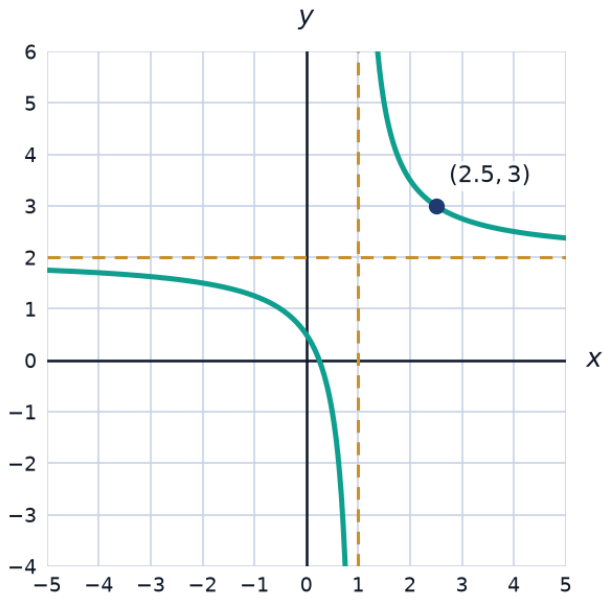
c. The graph crosses the asymptote

d. A corner is formed

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23 [Foundational] • Target: Multiplicity of Zeros and Sign Analysis

What is the sign behavior of $f(x) = \frac{1}{(x-2)^2}$ on both sides of $x = 2$?



a. Positive ($+\infty$) on both sides

b. Negative on left, positive on right

c. Negative on both sides

d. Undefined

24 [Foundational] • Target: Constructing Sign Charts

Which values partition the number line into test intervals when making a sign chart for a rational function?

a. All real zeros of the numerator and all real zeros of the denominator

b. Only the zeros of the denominator

c. Only the zeros of the numerator

d. Only the y-intercept

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25 [Foundational] • Target: Constructing Sign Charts

Find the solution set of $\frac{x-3}{x+1} > 0$.

a. $(-\infty, -1) \cup (3, \infty)$

b. $(-1, 3)$

c. $[3, \infty)$

d. $(-\infty, 3)$

26 [Foundational] • Target: Constructing Sign Charts

Solve the inequality $\frac{x^2-4}{x} \leq 0$.

a. $(-\infty, -2] \cup (0, 2]$

b. $[-2, 0) \cup [2, \infty)$

c. $[-2, 2]$

d. $(-\infty, 0)$

27 [Foundational] • Target: Sketching Complete Curves

Which of the following is the complete set of features needed to accurately sketch a rational function?

a. Domain, holes, vertical/horizontal asymptotes, x- and y-intercepts, and sign behavior

b. Only the degree and the leading coefficient

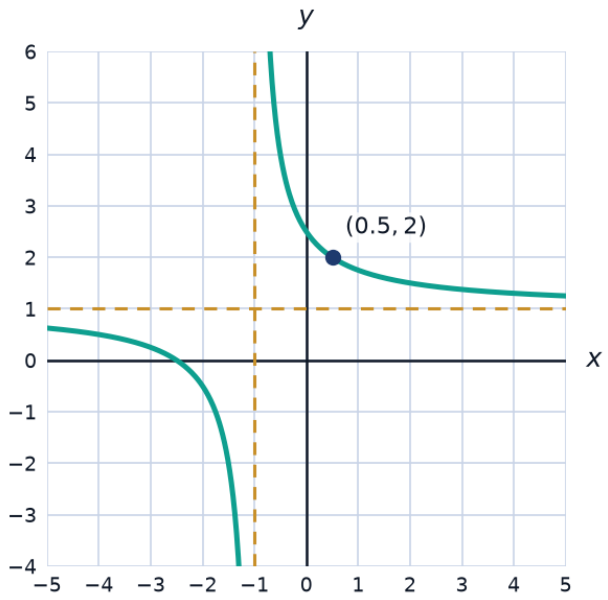
c. Only the y-intercept and one point

d. Only the vertical asymptotes

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28 [Foundational] • Target: Sketching Complete Curves

Sketch $f(x) = \frac{2x}{x-1}$. What is the value of $f(2)$ and $f(-1)$?



a. $f(2) = 4$ and $f(-1) = 1$

b. $f(2) = 2$ and $f(-1) = -1$

c. $f(2) = 4$ and $f(-1) = -1$

d. $f(2) = 1$ and $f(-1) = 2$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Sketching Complete Curves

In $f(x) = \frac{2x}{x-1}$, as $x \to 1^+$, what is the limit of $f(x)$?

a. $+\infty$

b. $-\infty$

c. 2

d. 0

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30 **[Intermediate]** • **Target: Writing Rational Functions from Features**

Write a rational function with vertical asymptotes at $x = -2$ and $x = 3$, horizontal asymptote $y = 0$, and x-intercept $(1, 0)$.

a. $f(x) = \frac{x - 1}{(x + 2)(x - 3)}$

b. $f(x) = \frac{(x + 2)(x - 3)}{x - 1}$

c. $f(x) = \frac{x - 1}{x^2 + 1}$

d. $f(x) = \frac{2x - 1}{(x + 2)(x - 3)}$

31 **[Intermediate]** • **Target: Writing Rational Functions from Features**

Write a rational function with vertical asymptote $x = 4$, hole at $x = -1$, horizontal asymptote $y = 2$, and y-intercept $(0, -1/2)$.

a. $f(x) = \frac{2(x + 1)(x - 1)}{(x + 1)(x - 4)}$

b. $f(x) = \frac{2x + 1}{x - 4}$

c. $f(x) = \frac{(x + 1)}{(x + 1)(x - 4)}$

d. $f(x) = \frac{2(x - 4)}{x + 1}$

32 **[Intermediate]** • **Target: Writing Rational Functions from Features**

What is the minimum degree of the denominator of a rational function with two distinct vertical asymptotes?

a. 2

b. 1

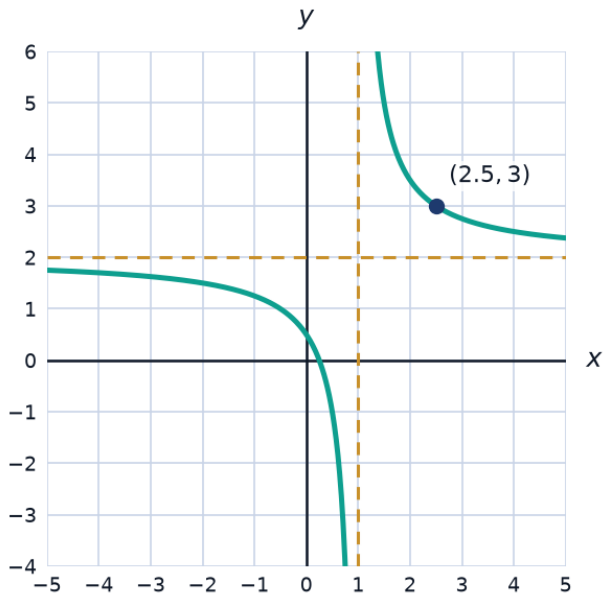
c. 3

d. 4

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33 [Intermediate] • Target: Writing Rational Functions from Features

For $f(x) = \frac{(x-4)(x+6)}{(x+6)(x+3)}$, identify the hole and vertical asymptote.



a. Hole at $x = -3$, vertical asymptote at $x = -6$

b. Vertical asymptotes at $x = -6$ and $x = -3$, no hole

c. Hole at $x = 4$, vertical asymptote at $x = -3$

d. Hole at $x = -6$, vertical asymptote at $x = -3$

34 [Intermediate] • Target: Reading graphs

For $f(x) = \frac{(x-5)(x+5)}{(x+5)(x+3)}$, identify the hole and vertical asymptote.

a. Hole at $x = 5$, vertical asymptote at $x = -3$

b. Hole at $x = -5$, vertical asymptote at $x = -3$

c. Hole at $x = -3$, vertical asymptote at $x = -5$

d. Vertical asymptotes at $x = -5$ and $x = -3$, no hole

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35 **[Intermediate]** • **Target: Sign analysis**

For $f(x) = \frac{(x-3)(x+2)}{(x+2)(x+4)}$, identify the hole and vertical asymptote.

a. Hole at $x = 3$, vertical asymptote at $x = -4$

b. Hole at $x = -4$, vertical asymptote at $x = -2$

c. Hole at $x = -2$, vertical asymptote at $x = -4$

d. Vertical asymptotes at $x = -2$ and $x = -4$, no hole

36 **[Intermediate]** • **Target: Sketching a complete graph**

For $f(x) = \frac{(x-5)(x+6)}{(x+6)(x+4)}$, identify the hole and vertical asymptote.

a. Hole at $x = 5$, vertical asymptote at $x = -4$

b. Hole at $x = -4$, vertical asymptote at $x = -6$

c. Hole at $x = -6$, vertical asymptote at $x = -4$

d. Vertical asymptotes at $x = -6$ and $x = -4$, no hole

37 **[Intermediate]** • **Target: More rational inequalities**

For $f(x) = \frac{(x-1)(x+6)}{(x+6)(x+4)}$, identify the hole and vertical asymptote.

a. Hole at $x = 1$, vertical asymptote at $x = -4$

b. Hole at $x = -6$, vertical asymptote at $x = -4$

c. Vertical asymptotes at $x = -6$ and $x = -4$, no hole

d. Hole at $x = -4$, vertical asymptote at $x = -6$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Building a rational function from a description • Cross-Lesson: 4.03 Hyperbolas

For $f(x) = \frac{(x+5)(x+3)}{(x+3)(x-0)}$, identify the hole and vertical asymptote.

a. Hole at $x = 0$, vertical asymptote at $x = -3$

b. Vertical asymptotes at $x = -3$ and $x = 0$, no hole

c. Hole at $x = -5$, vertical asymptote at $x = 0$

d. Hole at $x = -3$, vertical asymptote at $x = 0$

39 [Challenging] • Target: Graphing with a slant asymptote • Cross-Lesson: 5.01 Vectors: Components and Operations

For $f(x) = \frac{(x-5)(x+1)}{(x+1)(x-2)}$, identify the hole and vertical asymptote.

a. Vertical asymptotes at $x = -1$ and $x = 2$, no hole

b. Hole at $x = 5$, vertical asymptote at $x = 2$

c. Hole at $x = -1$, vertical asymptote at $x = 2$

d. Hole at $x = 2$, vertical asymptote at $x = -1$

40 [Challenging] • Target: Full graph analysis: putting it all together • Cross-Lesson: 5.02 The Dot Product

For $f(x) = \frac{(x+6)(x+2)}{(x+2)(x-0)}$, identify the hole and vertical asymptote.

a. Vertical asymptotes at $x = -2$ and $x = 0$, no hole

b. Hole at $x = -6$, vertical asymptote at $x = 0$

c. Hole at $x = -2$, vertical asymptote at $x = 0$

d. Hole at $x = 0$, vertical asymptote at $x = -2$

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41 [Challenging] • Target: A compound rational inequality • Cross-Lesson: 5.03 Parametric Equations

For $f(x) = \frac{(x-3)(x-0)}{(x-0)(x+3)}$, identify the hole and vertical asymptote.

a. Hole at $x = 0$, vertical asymptote at $x = -3$

b. Vertical asymptotes at $x = 0$ and $x = -3$, no hole

c. Hole at $x = 3$, vertical asymptote at $x = -3$

d. Hole at $x = -3$, vertical asymptote at $x = 0$

42 [Challenging] • Target: Graphing near a hole • Cross-Lesson: 6.01 Polar Coordinates and Conversions

For $f(x) = \frac{(x+6)(x-1)}{(x-1)(x-4)}$, identify the hole and vertical asymptote.

a. Hole at $x = 1$, vertical asymptote at $x = 4$

b. Hole at $x = -6$, vertical asymptote at $x = 4$

c. Vertical asymptotes at $x = 1$ and $x = 4$, no hole

d. Hole at $x = 4$, vertical asymptote at $x = 1$

43 [Challenging] • Target: Multiplicity of Zeros and Sign Analysis • Cross-Lesson: 6.02 Polar Equations and Graphs

For $f(x) = \frac{(x-1)(x-3)}{(x-3)(x-0)}$, identify the hole and vertical asymptote.

a. Hole at $x = 0$, vertical asymptote at $x = 3$

b. Hole at $x = 1$, vertical asymptote at $x = 0$

c. Vertical asymptotes at $x = 3$ and $x = 0$, no hole

d. Hole at $x = 3$, vertical asymptote at $x = 0$

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44 [Challenging] • Target: Constructing Sign Charts • Cross-Lesson: 7.01 Exponential Functions and Their Graphs

For $f(x) = \frac{(x-0)(x-3)}{(x-3)(x-5)}$, identify the hole and vertical asymptote.

a. Hole at $x = 5$, vertical asymptote at $x = 3$

b. Hole at $x = 0$, vertical asymptote at $x = 5$

c. Vertical asymptotes at $x = 3$ and $x = 5$, no hole

d. Hole at $x = 3$, vertical asymptote at $x = 5$

45 [Challenging] • Target: Sketching Complete Curves • Cross-Lesson: 7.02 Logarithms and Their Properties

For $f(x) = \frac{(x+4)(x-4)}{(x-4)(x-1)}$, identify the hole and vertical asymptote.

a. Hole at $x = 4$, vertical asymptote at $x = 1$

b. Vertical asymptotes at $x = 4$ and $x = 1$, no hole

c. Hole at $x = 1$, vertical asymptote at $x = 4$

d. Hole at $x = -4$, vertical asymptote at $x = 1$

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